

Practice Problems 5: Solutions

ECON 441: Introduction to Mathematical Economics

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Exercise 4.6

$$2. \ A = \begin{bmatrix} 0 & 4 \\ -1 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 3 & -8 \\ 0 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 & 9 \\ 6 & 1 & 1 \end{bmatrix}$$

$$(a) \ A + B = \begin{bmatrix} 3 & -4 \\ -1 & 4 \end{bmatrix}$$

$$A' + B' = \begin{bmatrix} 0 & -1 \\ 4 & 3 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ -8 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ -4 & 4 \end{bmatrix}$$

$$\text{So, } (A + B)' = A' + B'$$

$$(b) \ AC = \begin{bmatrix} 0 & 4 \\ -1 & 3 \end{bmatrix}_{2 \times 2} \begin{bmatrix} 1 & 0 & 9 \\ 6 & 1 & 1 \end{bmatrix}_{2 \times 3} = \begin{bmatrix} 24 & 4 & 4 \\ 17 & 3 & -6 \end{bmatrix}_{2 \times 3} \quad C'A' = \begin{bmatrix} 1 & 6 \\ 0 & 1 \\ 9 & 1 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 0 & -1 \\ 4 & 3 \end{bmatrix}_{2 \times 2} =$$

$$\begin{bmatrix} 24 & 17 \\ 4 & 3 \\ 4 & -6 \end{bmatrix}_{3 \times 2} \quad \text{So, } (AC)' = C'A'.$$

$$6. \ A = I - X(X'X)^{-1}X'$$

- (a.) Say the dimension of X is $m \times n$. Then the dimension of $X'_{n \times m} X_{m \times n}$ is $n \times n$. So the dimension of $(X'X)^{-1}$ is also $n \times n$. This implies that the dimension of $X_{m \times n} (X'X)^{-1}_{n \times n} X'_{n \times m}$ is $m \times m$. Hence, $X'X$ and A must be square matrices, but X need not be square.

(b.) To prove a matrix is idempotent, we need to show $AA = A$.

$$\begin{aligned} AA &= (I - X(X'X)^{-1}X')(I - X(X'X)^{-1}X') \\ &= I - X(X'X)^{-1}X' - X(X'X)^{-1}X' + \underbrace{X(X'X)^{-1}X'X(X'X)^{-1}X'}_I \\ &= I - X(X'X)^{-1}X' - X(X'X)^{-1}X' + X(X'X)^{-1}X' \\ &= I - X(X'X)^{-1}X' = A \end{aligned}$$